AMA 572: Credit Risk
Midterm Examination
Detailed Solutions

Detailed Solutions

Solution 1. Rating transitions and generator estimation

Solution.

The empirical estimator records each firm's state at the beginning and end of the year. Among the 12 firms initially in A, nine finish in A, two finish in B, and one defaults. Among the 8 firms initially in B, one finishes in A, six finish in B, and one defaults. Therefore,

$$\widehat{P}_{\text{emp}} = \begin{pmatrix} 9/12 & 2/12 & 1/12 \\ 1/8 & 6/8 & 1/8 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 0.7500 & 0.1667 & 0.0833 \\ 0.1250 & 0.7500 & 0.1250 \\ 0 & 0 & 1 \end{pmatrix}.$$

For state A, the original A firms contribute

$$9(1) + 2\left(\frac{3}{12}\right) + 1\left(\frac{9}{12}\right) = 10.25$$

firm-years. The firm that migrates from B to A at month 6 contributes another $6/12$. Thus,

$$N_A = 10.25 + 0.50 = 10.75 = \frac{43}{4}.$$

For state B, the original B firms contribute

$$6(1) + 2\left(\frac{6}{12}\right) = 7$$

firm-years. The two firms that migrate from A to B at month 3 contribute $2(9/12) = 1.5$ firm-years. Hence,

$$N_B = 7 + 1.5 = 8.5 = \frac{17}{2}.$$

The observed transition counts are

$$M_{AB} = 2, \quad M_{AD} = 1, \quad M_{BA} = 1, \quad M_{BD} = 1.$$

Consequently,

$$\widehat{\lambda}_{AB} = \frac{2}{43/4} = \frac{8}{43}, \qquad \widehat{\lambda}_{AD} = \frac{1}{43/4} = \frac{4}{43},$$

and

$$\widehat{\lambda}_{BA} = \widehat{\lambda}_{BD} = \frac{1}{17/2} = \frac{2}{17}.$$

The diagonal entries make each row sum to zero, so

$$\widehat{\Lambda} = \begin{pmatrix} -12/43 & 8/43 & 4/43 \\ 2/17 & -4/17 & 2/17 \\ 0 & 0 & 0 \end{pmatrix} \approx \begin{pmatrix} -0.27907 & 0.18605 & 0.09302 \\ 0.11765 & -0.23529 & 0.11765 \\ 0 & 0 & 0 \end{pmatrix}.$$

Exponentiating the matrix gives

$$\widehat{P}_{\text{gen}} = e^{\widehat{\Lambda}} \approx \begin{pmatrix} 0.76490 & 0.14439 & 0.09070 \\ 0.09131 & 0.79888 & 0.10981 \\ 0 & 0 & 1 \end{pmatrix}.$$

Unlike the direct empirical estimator, the generator estimator uses the exact amount of time spent in each state and allows multi-step migrations within the year.

Solution 2. Implied default probability from a bond price

Solution.

The corporate bond price implied by its yield is

$$B_{\text{corp}} = 104e^{-0.05} = 98.9279.$$

The otherwise identical default-free bond price is

$$B_{\text{rf}} = 104e^{-0.03} = 100.9263.$$

Thus, the price reduction attributed to default is

$$B_{\text{rf}} - B_{\text{corp}} = 1.9985.$$

At time 0.5, the risk-free value of the promised payment at time 1 is

$$104e^{-0.03(1-0.5)} = 104e^{-0.015} = 102.4516.$$

If default occurs, the recovery is 40, so the loss at time 0.5 is

$$102.4516 - 40 = 62.4516.$$

Its time-zero present value is

$$62.4516e^{-0.03(0.5)} = 61.5219.$$

Equating the bond price difference to expected discounted loss gives

$$1.9985 = p(61.5219),$$

and therefore

$$\boxed{p = 0.03248 \approx 3.25\%.}$$

Solution 3. Merton's structural model

Solution.

At maturity, equity holders receive the residual value after debt is paid:

$$S_T = (V_T - D)^+.$$

This is exactly the payoff of a European call option on V_T with strike D. Hence,

$$S_0 = V_0\Phi(d_1) - De^{-rT}\Phi(d_2),$$

where

$$d_1 = \frac{\log(V_0/D) + (r + \frac{1}{2}\sigma_V^2)T}{\sigma_V\sqrt{T}}, \qquad d_2 = d_1 - \sigma_V\sqrt{T}.$$

Numerically,

$$d_1 = \frac{\log(100/80) + (0.05 + 0.5(0.30)^2)}{0.30} = 1.06048,$$

and

$$d_2 = 1.06048 - 0.30 = 0.76048.$$

Using $\Phi(d_1) = 0.85554$ and $\Phi(d_2) = 0.77652$,

$$\begin{aligned} S_0 &= 100(0.85554) - 80e^{-0.05}(0.77652) \\ &= 26.4621. \end{aligned}$$

With no taxes or bankruptcy costs, debt plus equity equals firm value. Thus,

$$B_0 = V_0 - S_0 = 100 - 26.4621 = 73.5379.$$

Default occurs at maturity when $V_T < D$. Therefore,

$$\mathbb{P}^*(V_T < D) = \Phi(-d_2) = \Phi(-0.76048) = 0.22348.$$

Thus, the risk-neutral one-year default probability is approximately 22.35%.

The continuously compounded risky debt yield y satisfies

$$B_0 = De^{-yT}.$$

Hence,

$$y = \frac{1}{T}\log\left(\frac{D}{B_0}\right) = \log\left(\frac{80}{73.5379}\right) = 0.08423.$$

The credit spread is

$$\boxed{y - r = 0.08423 - 0.05 = 0.03423 \approx 3.42\%,}$$

or about 342 basis points.

Solution 4. First passage time and the Black-Cox model

Solution.

Taking logarithms of the asset process gives

$$\log\left(\frac{V_t}{V_0}\right) = \mu t + \sigma W_t.$$

Thus, τ is the first time a Brownian motion with drift μ and volatility σ hits the lower level $b < 0$. Its Laplace transform is

$$\boxed{\mathbb{E}[e^{-s\tau}] = \exp\left\{\frac{b}{\sigma^2}\left(\sqrt{\mu^2 + 2\sigma^2 s} + \mu\right)\right\}, \qquad s > 0.}$$

The reflection principle for Brownian motion with drift gives the probability that the running minimum crosses b by time T:

$$\boxed{\mathbb{P}(\tau \le T) = \Phi\left(\frac{b - \mu T}{\sigma\sqrt{T}}\right) + \exp\left(\frac{2\mu b}{\sigma^2}\right)\Phi\left(\frac{b + \mu T}{\sigma\sqrt{T}}\right).}$$

The two terms correspond to paths ending below the barrier and paths ending above the barrier after having crossed it earlier.

For the stated parameters,

$$\mu = 0.04 - 0.02 - \frac{1}{2}(0.25)^2 = -0.01125, \qquad b = \log(0.70) = -0.356675.$$

Also,

$$\frac{b - \mu T}{\sigma\sqrt{T}} = -0.94519, \qquad \frac{b + \mu T}{\sigma\sqrt{T}} = -1.07247,$$

and

$$\exp\left(\frac{2\mu b}{\sigma^2}\right) = 1.13701.$$

Therefore,

$$\begin{aligned}\mathbb{P}(\tau \le 2) &= \Phi(-0.94519) + 1.13701\Phi(-1.07247) \\ &= 0.17228 + 1.13701(0.14175) \\ &= 0.33346.\end{aligned}$$

Thus,

$$\boxed{\mathbb{P}(\tau \le 2) \approx 33.35\%.}$$

Solution 5. Leland-Toft endogenous default

Solution.

At time t, debt issued at time $u < t$ has remaining maturity $s - t$ when its original maturity is $s - u$. The par value maturing in $(s, s + \mathrm{d}s)$ is

$$\left(\int_{-\infty}^{t} pme^{-m(s-u)} \, \mathrm{d}u \right) \mathrm{d}s = pe^{-m(s-t)} \, \mathrm{d}s.$$

Integrating over all future maturity dates gives

$$P = \int_{t}^{\infty} pe^{-m(s-t)} \, \mathrm{d}s = \frac{p}{m}.$$

At default, the aggregate present value of the individual bond recoveries is

$$\begin{aligned} \int_{\tau}^{\infty} ce^{-r(s-\tau)} pe^{-m(s-\tau)} \, \mathrm{d}s &= cp \int_{0}^{\infty} e^{-(m+r)u} \, \mathrm{d}u \\ &= \frac{cp}{m+r} = \frac{cmP}{m+r}, \end{aligned}$$

where $p = mP$ was used. Matching this amount to αV_B yields

$$\frac{cmP}{m+r} = \alpha V_B, \qquad \boxed{c = \frac{m+r}{m} \frac{\alpha V_B}{P}.}$$

The Modigliani-Miller decomposition in this setting is

$$v(V; V_B) = V + \mathbb{E}\left[\int_{0}^{\tau} \kappa \rho P e^{-rt} \, \mathrm{d}t \right] - (1 - \alpha) V_B \mathbb{E}[e^{-r\tau}].$$

Since

$$\int_{0}^{\tau} e^{-rt} \, \mathrm{d}t = \frac{1 - e^{-r\tau}}{r},$$

we obtain

$$\begin{aligned} v(V; V_B) &= V + \frac{\kappa \rho P}{r} \left(1 - \mathbb{E}[e^{-r\tau}] \right) - (1 - \alpha) V_B \mathbb{E}[e^{-r\tau}] \\ &= V + \frac{\kappa \rho}{r} P - \left[\frac{\kappa \rho}{r} P + (1 - \alpha) V_B \right] \mathbb{E}[e^{-r\tau}]. \end{aligned}$$

Using

$$\mathbb{E}[e^{-r\tau}] = \left(\frac{V_B}{V} \right)^{\gamma_r}, \qquad \gamma_r = \sqrt{\xi^2 + \frac{2r}{\sigma^2}} + \xi,$$

gives

$$\boxed{v(V; V_B) = V + \frac{\kappa \rho}{r} P - \left[\frac{\kappa \rho}{r} P + (1 - \alpha) V_B \right] \left(\frac{V_B}{V} \right)^{\gamma_r}.}$$

The optimization has two stages because debt and equity holders have different objectives. For a fixed debt level P, equity holders choose the default boundary V_B to maximize equity value subject to limited liability. The value-matching and smooth-pasting conditions at the boundary lead to a boundary of the form $V_B^*(P) = \varepsilon P$. At the second stage, the firm anticipates this equity-holder decision and chooses P to maximize $v(V; V_B^*(P))$. The optimum balances the tax advantage of debt against expected bankruptcy costs.

Solution 6. Duffie-Singleton reduced-form pricing

Solution.

Over a short interval $\mathrm{d}t$, default occurs with probability approximately $\lambda\,\mathrm{d}t$. If default occurs, the holder keeps the fraction R_M of market value; otherwise, the full continuation value is retained. Ignoring terms of order $(\mathrm{d}t)^2$, the one-step discount multiplier is

$$\begin{aligned} e^{-r\,\mathrm{d}t}\left[\lambda\,\mathrm{d}t\,R_M + (1-\lambda\,\mathrm{d}t)\right] &\approx (1-r\,\mathrm{d}t)\left[1-(1-R_M)\lambda\,\mathrm{d}t\right] \\ &\approx 1-\left[r+(1-R_M)\lambda\right]\,\mathrm{d}t. \end{aligned}$$

Thus, the effective discount rate is

$$\boxed{r+(1-R_M)\lambda.}$$

For a zero-coupon bond with unit face value,

$$B(0,T) = \exp(-[r+(1-R_M)\lambda]T)\,.$$

With $T=4$, $r=0.03$, $\lambda=0.02$, and $R_M=0.40$,

$$r+(1-R_M)\lambda = 0.03+0.60(0.02) = 0.042.$$

Therefore,

$$\boxed{B(0,4) = e^{-0.042(4)} = e^{-0.168} = 0.84535.}$$

The continuously compounded yield is 4.20%, and the credit spread is

$$\boxed{4.20\% - 3.00\% = 1.20\% = 120 \text{ basis points.}}$$

Solution 7. Five-year CDS valuation

Solution.

Let

$$S_k = (1-q)^k$$

be the probability of survival through year k, and let

$$Q_k = q(1-q)^{k-1}$$

be the probability of default during year k. The regular premium coefficient at time k is $S_k e^{-rk}$. The accrued-premium coefficient is $\frac{1}{2}Q_k e^{-r(k-1/2)}$, and the protection-leg contribution is $(1-R)Q_k e^{-r(k-1/2)}$.

k	S_k	Q_k	$S_k e^{-rk}$	$\frac{1}{2}Q_k e^{-r(k-1/2)}$	$(1-R)Q_k e^{-r(k-1/2)}$
1	0.975000	0.025000	0.936770	0.012252	0.015928
2	0.950625	0.024375	0.877537	0.011478	0.014921
3	0.926859	0.023766	0.822051	0.010752	0.013978
4	0.903688	0.023171	0.770072	0.010072	0.013094
5	0.881096	0.022592	0.721380	0.009435	0.012266
		Total	4.127810	0.053990	0.070187

Therefore, the present value of the premium leg is

$$\mathrm{PV}_{\mathrm{prem}} = s(4.127810 + 0.053990) = 4.181800s.$$

The present value of the protection leg is

$$\mathrm{PV}_{\mathrm{prot}} = 0.070187.$$

At the break-even spread these values are equal, so

$$s = \frac{0.070187}{4.181800} = 0.0167838.$$

Hence,

$$\boxed{s \approx 1.6784\% = 167.84 \text{ basis points per year.}}$$

End of solutions.